

Ariel K. Tran - Writing Sample

The following writing sample is my capstone for my M.S. in Computational Social Science. This capstone project was produced under the partnership of UC San Diego and the San Diego Hunger Coalition, a non-profit that performs data-driven research on hunger in the San Diego area to report to local hunger related community partners and governing bodies.

This report was edited to focus on my own writing and direct contributions, as the full version is the work of all three students in my capstone group.

**A Series of Methodological Updates to the SDHC Nutrition
Insecurity Model:
Evaluating Self-Purchased Meal Calculations with State-Level
Survey Weighting**

Ariel Tran

Capstone Host: Joseph Kendrick

Advisor: Aram Grigoryan

University of California San Diego

In partnership with the San Diego Hunger Coalition

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Introduction

Nutrition insecurity remains at the forefront of humanistic emergencies, and San Diego County is no exception. At least 860,000 (26%) of residents in San Diego County are estimated to be unable to obtain three nutritious meals a day, as of June 2025. The San Diego Hunger Coalition (SDHC) is an organization committed to remediating this crisis by creating equitable access to food assistance in San Diego County via research, education, and advocacy. SDHC monitors nutrition insecurity through a data-driven methodology integrating census data, food assistance program records, and economic indicators. Despite its scope, the existing methodology (A) relies on national-level estimates that do not reflect California's cost of living, (B) lacks demographic stratification at the sub-county level, and (C) operates on reporting lags of nine or more months that limit its usefulness for timely decision-making. This capstone project, conducted in partnership with SDHC, proposes and implements three methodological updates to address these gaps.

My methodological contribution is as follows:

(A) California state-weighted self-purchased meal estimates.

SDHC's current methodology applies unweighted national averages from the Consumer Expenditure Survey to estimate self-purchased meals by income bracket. We apply California-specific survey weights for 2016-2023, in hopes of producing estimates better calibrated to the state's economic conditions. California household food expenditure is modestly higher than national averages across most income brackets. Unfortunately, California-specific survey weights were recently discontinued with the latest release of 2024 data. As such, we recommend unweighted averages, nationally-weighted averages, and medians as central estimation values for self-purchased meal estimates. These three recommendations depend on the organization's use cases and estimation priorities.

Background

One of the central ways SDHC approximates nutrition insecurity is by calculating the meal gap. This is estimated by taking the total meals needed, three nutrition meals per day, subtracting self-purchased meals and total food assistance. Current methodology for estimating self-purchased meals comes from the Interview Survey under the Consumer Expenditure Survey (CEX) by the Bureau of Labor Statistics (BLS) and US Census Bureau. On a quarterly basis, these households across the United States are surveyed on income, SNAP issuance, national level sample weights, demographic details, and major expenditures such as "food at home". This variable measures grocery expenditure and excludes money spent on travel or dining related food. As such, when measuring the meal gap, self-purchased meals (SPM) is taking "food at home" expenditure and subtracting CalFresh to avoid double counting Calfresh again under total food assistance. This effectively measures raw consumer grocery expenditure.

SPM averages are stratified across federal poverty levels (FPL), income brackets used to define eligibility levels for various federal, state, non-profit, and welfare programs (HSS citation). The U.S. Department of Health and Human Services (HHS) sets the guidelines for the poverty line, defined as the minimum income to cover basic needs, and is referred to as the 100% federal poverty level. This measurement system is set up to be expansive across different household sizes and distance to the poverty line (see Table 1). For example, the 200% FPL is twice as much as the poverty line at 100%. Hence, the income for a family size of 1 at 200% FPL is twice as much for the same family size at 100% FPL. This multiplicative relationship extends to increase in dollars for each additional family member across households of the same FPL. For example, a household size of 5 at the 150% FPL has an income that is at least \$8,250 greater than a household size of 4. Therefore, a household size of 5 has an income of at least \$48,225 + \$8250 = \$56,475. FPL is multiplicative with percentages across different levels and additive with dollars for each additional family member.

2022 Federal Poverty Level (FPL)					
Family Size	0%	50%	100%	150%	200%
1	\$0	\$6,795	\$13,590	\$20,385	\$27,180
2	\$0	\$9,155	\$18,310	\$27,465	\$36,620
3	\$0	\$11,515	\$23,030	\$34,545	\$46,060
4+	0	Add \$2,750 for each additional family member	Add \$5,500 for each additional family member	Add \$8,250 for each additional family member	Add \$10,100 for each additional family member

Table 1.

To further contextualize FPLs, an example household size of 3 in the metropolitan area of San-Diego-Chula-Vista-Carlsbad is considered working class at approximately 0-250% FPL, middle class around 300-550%, and upper class beyond 550% (Fry 2023). SDHC's estimations based on 2023 CEX data have found that even people at 225% FPL are not spending enough to afford 3 healthy meals per day.

The CEX has been releasing national-level sample weights from 20XX to 20XX and state-level weights from 2016-2023 for four states including California. These weights are uniquely calculated for each household taking into account nonresponse rates and geography. Samples weights can range from around 3,000 to 160,000. Interpretation examples are listed below.

- Example 1) A national-level weight value of 38,380 means that the responses of a sampled household is estimated to represent 38,380 real households across the US.
- Example 2) A California state-level weight value of 13,489 means that the responses of a sampled household is estimated to represent 13,489 real households across California.

Current SDHC self-purchase meal estimations are un-weighted and national-level averages across FPLs. Therefore, applying California level weights and US level weights will provide new estimates. Comparing the prior estimates after changing the geographic scope and applying survey weights could reveal shortcomings related to the relevancy and validity of the estimation method.

It is important to note that since the initial release of state weights for CEX data, state weights have been noted as a “research product” and “may not be consistently available across the five listed states going forwards as they are highly dependent on sample composition” (BLS 2024). Additionally, the latest release of data (2024) was the first year since state weights were first released in which state weights were not released. After emailing an official BLS representative about no 2024 weights, they confirmed that as of 2-6-2026, “Unfortunately, at this time there are no plans to continue producing State Weights.” Therefore, using this geographic and weight data is effectively a historical evaluation of the current methodology and historical analysis of SPM expenditure.

Adjusting self-purchased meal estimates to a state level will be crucial due to California’s economy and higher cost of living. California is the largest importer and second-largest exporter in the nation (Payares-Montoya, 2025). As of February 2025, many of California’s biggest trading partners, Mexico, China, and Canada, have tariffs between 25%-35% (Avalara, 2025). This and reciprocal tariffs has and will continue to affect Californian’s ability to purchase food. Therefore, adjusting the nationally weighted Consumer Expenditure Survey data with CA state weights is increasingly important for evaluating and monitoring self-purchased meals under this state specific and changing economy.

Research Question

RQ: With the sudden discontinuation of California-level weights for grocery expenditure census data, how do these weights inform our understanding of: (1) California-specific grocery spending estimates and (2) evaluation of San Diego Hunger Coalition’s current grocery estimation methodology?

Data Preparation

Because there are only CA-level weights for 2016 to 2023 data, analysis focused on the same years worth of data. Eight years of data from the Interview Survey, CEX weights, and Federal Poverty Levels were merged. This dataset was subsetting to only those at or below 225% FPL, those who are not spending enough for three nutrition meals per day. Because 0-250% FPL

is considered working class income for the San-Diego-Chula-Vista-Carlsbad area, this subsetting data represents many working class households.

Households sampled from California had a unique CA-level weight, and non-Californian ones did not have a state weight. All households had a national-level weight. The Interview Survey resamples the same household up to four consecutive quarters in a row. A household's reported quarterly food expenditure was divided four and by the household's size to get a more interpretable expenditure variable on a monthly and per person basis. Yearly income was divided by twelve and the household size for monthly and individual level numbers.

The resulting national dataset featured 50,807 households and 25,192 unique households. This subsetting data covers the bottom 38.41% of households in terms of FPLs. The Californian subset dataset covers 5,024 rows and 2,490 unique households. Due to the natural attrition of resampling the same households, the average appearances of a unique household was 2.02. Longitudinal analysis will have repeated households that appear between one to four times.

Exploratory Analysis

With a central focus on exploring how SPM estimates vary with CA-level weights, it's important to establish how much pre-weighting SPM varies between CA and US households and the distribution of FPLs. Figure 1 shows that surveyed households are distributed with a left skew across FPLs. This is expected as lower income households are historically more likely to decline government surveying. This lower sampling in lower FPLs highlights the dataset's underrepresentation of the most nutritionally vulnerable populations.

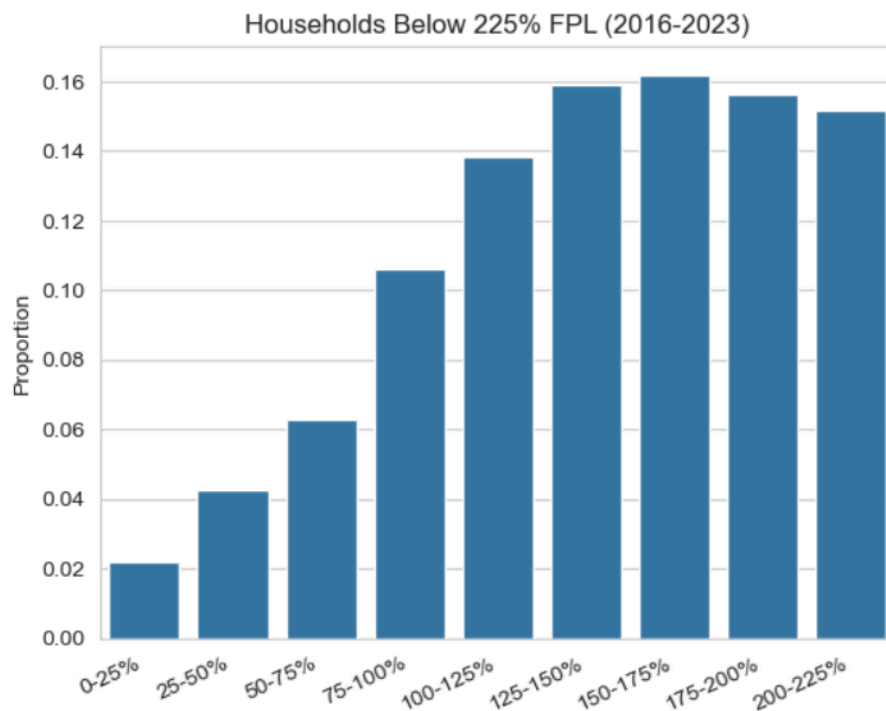


Figure 1.

When observing SPM expenditure over time across different FPLs, the overall trend of an increasingly higher SPM expenditure post-pandemic 2020 and beyond is consistent with the Consumer Price Index on Food at Home. Essentially, the unweighted data for US and CA households generally aligns with inflation trends. Comparing longitudinal food expenditure between the US and CA highlights how the CA-subsetting leads to much more noise and varied aggregate spending (Figure 2). This noise is consistent with the fact that food spending is very personal and the CA subsetting data represent only 9.89% of the full national data.

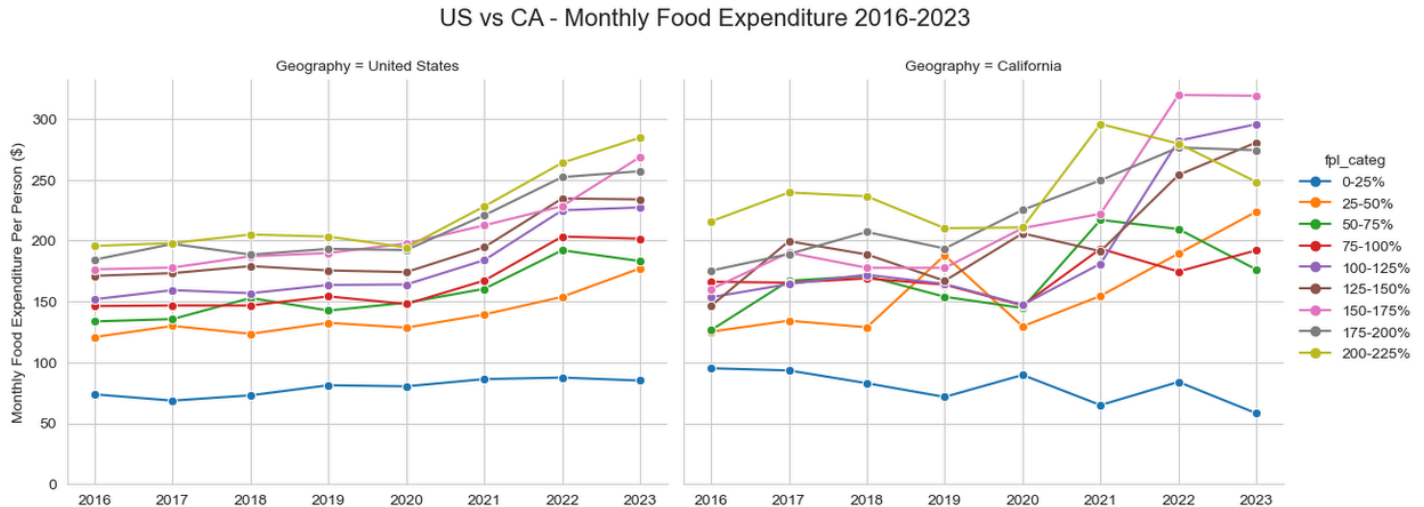


Figure 2.

SPM expenditure averages across FPLs for households in California are only slightly higher than those across the U (Figure 3). This relationship is seen across the majority of FPL categories under 225%. Each grey dot on this chart represents the amount spent on food per person on a monthly basis for a different household aggregated across FPLs. The data points are more concentrated below and around the averages. The gradual graduation of dots upwards on food expenditure emphasizes that food spending is greatly varied for most households and FPLs.

The consistent but small difference in average food spending between US and CA households continues when examining SPM propensity scores, the proportion of income spent on SPMs (Figure 4). Average food propensity for CA households is generally slightly higher than US households. Figure 4 also shows sharper decline in food propensity from 0% to 100% FPL before it starts to plateau. This reemphasizes the unfortunate reality that many households below the federal poverty line must spend a relatively disproportionate large portion of their income on food, a basic necessity. Figure 3 and 4 establish the consistent yet small difference in average food spending behavior across nutrition insecure populations when comparing households across CA and the US.

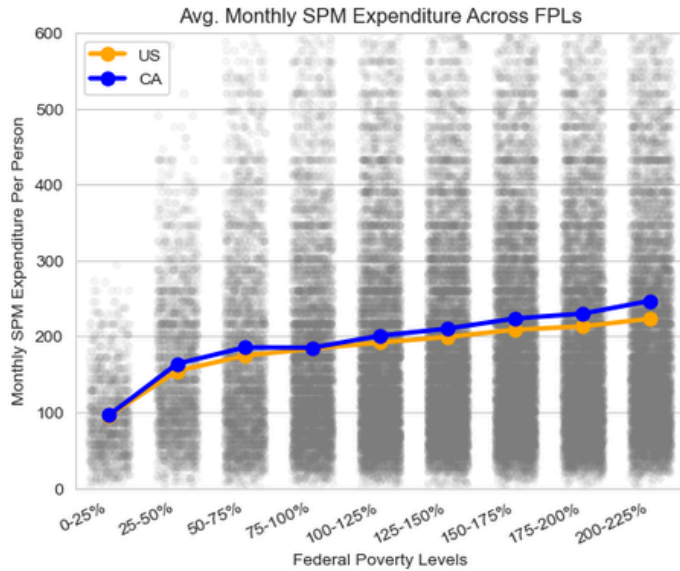


Figure 3.

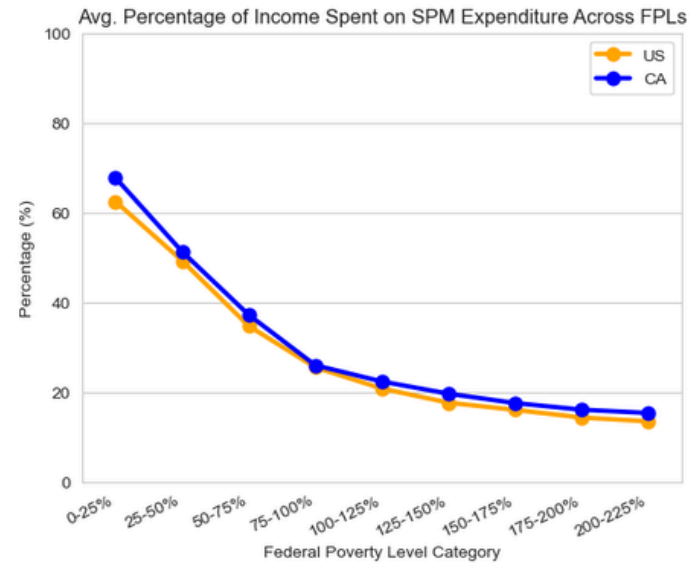


Figure 4.

Statistical Analysis

To test if the difference in SPM expenditure between CA and US households is statistically significant whilst controlling for income, geographic location, SNAP issuance, and inflation, a multivariate regression was conducted separately for CA and US data. Income, snap issuance, and food expenditure are right skewed and were logged in order to uphold OLS assumptions of normally distributed residuals. Model outputs and differences can be seen in Table 2. All model outputs were statistically significant with a p-value below 0.001. Compared to the US households, CA households on average have a slightly lower intercept, higher return on income, lower return on SNAP, and higher return quarter to quarter.

Therefore, the difference in CA and US average SPM spending behavior is statistically significant after controlling for SNAP issuance, income, and inflation. However, when taking the absolute value of all coefficients, they are all below 0.3. This means that the rate of change is 0.004-3.000% across all parameters. As such, there is a statistically significant but relatively small difference in at-home food spending behavior when comparing households across CA to across the nation.

Predicting Monthly SPM Expenditure (ln) Per Person with Multivariate Models		
	US Model	CA Model
Intercept	0.825*** (0.019)	0.754*** (0.062)

Monthly SNAP Issuance Per Person (ln)	-0.192*** (0.003)	-0.201*** (0.009)
Monthly Income Per Person (ln)	0.185*** (0.003)	0.197*** (0.009)
Quarter (¼ of a Calendar Year)	0.004*** (0.000)	0.005*** (0.001)
Observations	50,807	5,024
R2	0.210	0.205
Adjusted R2	0.210	0.204
Residual Std.Error	0.358 (df=50803)	0.364 (df=5020)
F Statistics	4509.403*** (df=3; 50803)	431.323*** (df=3; 5020)
Note: *p<0.1. **p<0.05; ***p<0.01		

Table 2.

Applying Weights

To compare raw averages and weighted averages, US weights were applied onto all households and CA weights were applied to the CA subset of households. After calculating the weighted average SPM expenditure across FPLs, side-by-side plots below show that CA households on average have higher spending pre and post weighting. The overall shape and curvature of the lines are similar with and without weights, highlighting that CA and US weights did not dramatically up or down weight any particular FPL range relative to each other. Overall, weighted averages were slightly less than their unweighted averages with the largest difference being approximately \$20 (6% greater).

Logistical Concerns with Weight Data

Even if state weights were not discontinued, there were multiple other shortcomings. State weight documentation notes that this estimation method cannot produce valid weighted variation measures such as weighted standard deviation or weight standard error. Weighted averages would be missing crucial contextual information from weighted variance calculations.

Additionally, stratified state-level averages would lead to sample sizes that are too small. Table 3 highlights that sample sizes once stratified across FPLs are consistently small, especially with sample sizes ranging between 4 to 46 for the 0-50% FPLs. In contrast, the national dataset has sample sizes ranging between 135 to 558 for the 0-50% FPLs. Surveyed sample sizes for the

most financially vulnerable populations are expectedly small. Subsetting to just households in California and across income categories stratifies sample sizes too much. Even without weighted standard error estimates, the sample size produces weighted average values that are too unreliable and risk poor generalizability. Because stratified estimates are essential to SDHC meal gap calculations, subsetting the national level data from CEX Survey proves unfit for California level estimates whether weighted or not.

Stratified Sample Sizes Across FPLS for California Data								
FPL	2016	2017	2018	2019	2020	2021	2022	2023
0-25%	10	14	12	8	20	9	13	4
25-50%	46	36	34	22	17	19	28	19
50-75%	60	58	57	47	41	33	37	24
75-100%	98	106	94	71	78	51	69	59
100-125%	128	120	97	87	72	75	62	72
125-150%	105	124	102	98	88	72	88	80
150-175%	114	118	113	81	81	90	73	88
175-200%	111	101	111	82	82	100	96	92
200-225%	108	98	113	87	67	84	93	81

Table 3. Heat map of sample sizes stratified across FPLs. Lighter (white) colors indicate smaller and more vivid (orange) colors indicate larger sample size.

Comparing Central Tendency Values

For both the US-level and CA-level data, the weighted means are modestly less than their respective unweighted means (Figure 5). The medians are less than their respective unweighted and weighted means. These relationships are consistent from 50-225% FPL. Again, as shown in previous figures, the weighted means and unweighted means between the US and CA data are very close, with CA modestly higher.

Median food expenditure is lower because medians are more resistant to the right skewed variability in the data. This means the above average expenditure has less of an effect on the central tendency value.

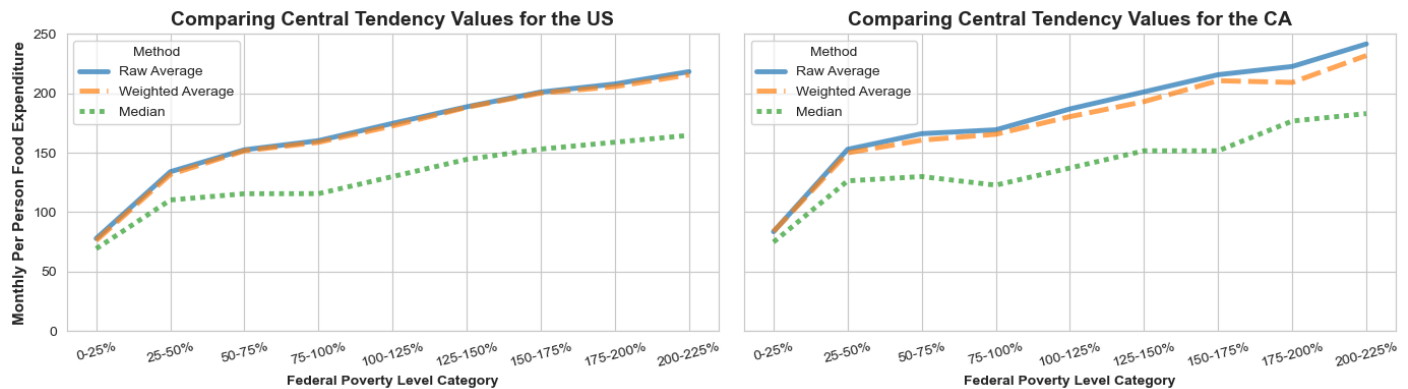


Figure 5.

Conclusion and Recommendations

Historical analysis of SPM expenditure for the population at risk of hunger, those at or below 225% FPL, reveals that households in California are spending more on food compared to the national averages. This is generally consistent across FPLs and in both weighted and unweighted estimates. After controlling for SNAP issuance, income, and time, California food spending is still greater than national-level food spending on average with statistical significance. This established that CEX data is consistent with the higher cost of living in California.

The state-level weights offered the opportunity to subset only Californian households and to apply weights that were more localized than national-level weights. Modest estimation differences were found after comparing SPM estimates across national-level weights, CA-level weights, all national households, and California-only households. Because state weight data was recently discontinued, intentions of localizing SPM estimates through the CEX Survey are not possible. The state weight data is the only variable that contains geographic data denoting which households are from California. Therefore, unweighted averages of California only households are also discontinued. At best, exploring how estimates vary suggests that SDHC's current unweighted and national-level means are relatively similar to CA-weighted averages and CA unweighted averages.

Taking into account the logistical concerns with state weights, SPM estimation recommendations include continuing with un-weighted national averages or nationally weighted averages depending on the value of continuous methodology. Un-weighted national averages means that SDHC's estimates will be readily consistent with prior published estimates. This continuity poses the least logistical concerns, as SDHC estimates affect several other nutrition organizations and local governments. Seeing what estimates would have been with weighting methodology suggests that pre and post weight averages are more similar than expected. The alternative recommendation is to produce national level and nationally weighted estimates. This method is viable as state weights were discontinued and national weights have been consistently released since 1984. Because nationally weighted estimates were very similar to California

weighted estimates, this method would be the closest and more logistically available way to update self-purchased meal estimates. The last recommendation would be to use medians as this would be more resistant to the right skewed variation of food spending behavior. Being more resistant to higher food spending within a federal poverty level means representing households with a lower food spending, those who are more likely to be experiencing stronger food insecurity. The downside of medians is that this would be the most abrupt change to self-purchased meal estimates. While medians better represent the nutrition insecure population that is not meeting the standard of being able to afford three healthy meals per day, median estimation is more inconsistent.

To further explore these three estimation recommendations, applying different self-purchased meal estimations to the meal gap calculation would allow the San Diego Hunger Coalition to better assess these alternative methods.

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